According to Gartner, the cloud computing market is projected to be worth US$ 441 billion by 2020. Microsoft Azure and Amazon Web Services are also investing heavily in edge computing.

AWS DocumentDB service launch triggers controversy

Tension between the open source community and cloud platforms resurfaced recently after Amazon Web Services (AWS) announced a new managed document database service, which uses open source MongoDB code.

Amazon DocumentDB is described by the company as a fast and scalable document database that is designed to be compatible with existing MongoDB apps and tools.

Soon after the AWS DocumentDB service was announced, complaints and objections started to pour in, with many alleging the cloud major of infringing upon open source projects.

CNBC.com quoted MongoDB CEO Dev Ittycheria as saying: "Imitation is the sincerest form of flattery, so it's not surprising that Amazon would try to capitalise on the popularity and momentum of MongoDB. However, developers are savvy enough to distinguish between the real thing and a poor imitation."

Tweeters also erupted in anger with posts such as: "This is the #MongoDB service. Not even API compatible. Almost a copy."

In order to stop large vendors from exploiting its freely available technology, MongoDB had recently released a set of public licensing policies for third-party commercial use. The licence, however, applies to newer editions of the database. AWS apparently uses older MongoDB code in its service.

Amazon DocumentDB uses a purpose-built SSD-based storage layer, with six-way replication across three availability zones. "The storage layer is distributed, fault-tolerant and self-healing, giving you the performance, scalability and availability needed to run production-scale MongoDB workloads," AWS chief evangelist Jeff Barr says in a blog post.

With Amazon DocumentDB, the company claims storage can be scaled from 10GB up to 64TB in increments of 10GB. It stores database changes as a log stream, allowing users to process millions of reads per second with millisecond latency. Meanwhile, the six-way storage replication ensures high availability.

Like the other AWS database services, Amazon DocumentDB is fully managed, with built-in monitoring, fault detection and failover. "Amazon DocumentDB is compatible with version 3.6 of MongoDB. Internally, it implements the MongoDB 3.6 API by emulating the responses that a MongoDB client expects from a MongoDB server," Barr says.

Amazon DocumentDB is available now for use in the US East (North Virginia and Ohio) and US West (Oregon) regions, as well as in Ireland.

"Pricing is based on the instance class, storage consumption for current documents and snapshots, I/O operations, and data transfer," Barr adds.

Ockam open sources the code of its IoT SDK

Ockam, an IoT development platform provider, has announced the release of its core software development kit (SDK) on GitHub.

Software that lets anyone create 2D nanostructures using DNA strands

A team of scientists from MIT and Arizona State University has designed a computer program that allows users to translate any free-form drawing into a two-dimensional, nano-scale structure made of DNA. The program is called PERDIX (Programmed Eulerian Routing for DNA Designs using X-overs).

Using this new program, anyone can create a DNA nano-structure of any shape, for applications in cell biology, photonics, quantum sensing and computing, among many others, the team said.

"What this work does is allow anyone to draw literally any 2D shape and convert it into DNA origami automatically," says Mark Bathe, an associate professor of biological engineering at MIT and the senior author of the study.

The new technology is built using the principle of scaffolded DNA origami, invented by Dr Paul Rothemund in 2006.

PERDIX automatically converts any 2D polygonal mesh file specified using simple computer-aided design (CAD) into the synthetic DNA sequences needed to synthesise the target object. In short, PERDIX offers the ability to 'print' 2D nanometer-scale geometry using DNA.

The shape can be sketched in any computer drawing program and then converted into a computer-aided design (CAD) file, which is fed into the DNA design program. "Once you have that file, everything's automatic, much like printing, but here the ink is DNA," Bathe explains.

"The goal of PERDIX is to cross disciplines and democratise the usage of structured DNA assemblies by offering a fully autonomous, CAD geometry-based sequence design algorithm. We hope that you can use PERDIX to explore the capabilities of the powerful molecular design paradigm of scaffolded DNA origami for applications of your own interest," the MIT team says.

PERDIX is freely available under the GNU General Public License, version 3 (GPL-3.0). Source code is available in Fortran and can be downloaded via GitHub.